DFO, GDPopt and MINLP completion times — all nine models

Each cell: completion time / target attainment count. All 45 GDPopt and MINLP jobs are included.

Model	DSDA	RBFOpt	BAM 5.0.0	FMQA-SA	RIC GDPopt	GLOA GDPopt	LOA GDPopt	Big-M MINLP	Hull MINLP
CSTR 5 DFO B=8	21.72 s 3/10	12.73 s 10/10	24.80 s 10/10	20.01 s 2/10	31.27 s 1/1	3.76 s 0/1	31.82 s 1/1	9.12 s 1/1	14.86 s † 1/1
CSTR 10 DFO B=15	12.52 s 3/10	18.60 s 7/10	11.83 s 1/10	13.57 s 2/10	214.06 s 1/1	4.56 s 0/1	234.13 s 1/1	123.74 s 1/1	126.47 s † 1/1
CSTR 15 DFO B=22	29.21 s 2/10	48.99 s 10/10	19.50 s 1/10	34.21 s 1/10	1798.83 s TL 0/1	26.19 s 0/1	1799.38 s TL 1/1	1799.00 s TL 1/1	940.36 s † 1/1
CSTR 20 DFO B=29	27.47 s 3/10	53.70 s 9/10	31.50 s 2/10	40.80 s 1/10	1799.04 s TL 0/1	6.79 s 0/1	1797.72 s TL 0/1	1798.72 s TL 1/1	1797.44 s TL 1/1
CSTR 25 DFO B=36	46.30 s 3/10	99.81 s 10/10	46.12 s 2/10	52.18 s 1/10	1797.05 s TL 0/1	85.20 s 0/1	1797.46 s TL 0/1	1797.67 s TL 1/1	1797.41 s TL 1/1
CSTR 30 DFO B=43	51.13 s 5/10	97.91 s 10/10	33.87 s 2/10	65.06 s 2/10	1792.52 s TL 0/1	718.80 s 0/1	1798.92 s TL 0/1	1797.31 s TL 1/1	1466.60 s † 0/1
Column DFO B=16	46.20 s 7/9	43.21 s 4/9	40.62 s 0/9	28.97 s 4/9	1797.54 s TL 0/1	1798.15 s TL 0/1	1797.77 s TL 0/1	34.36 s † 0/1	1797.17 s TL 0/1
Small Batch DFO B=10	7.94 s 10/10	8.78 s 10/10	8.04 s 10/10	8.36 s 9/10	12.86 s 1/1	14.49 s 1/1	5.11 s 1/1	0.55 s 1/1	0.57 s 1/1
Dynamic DFO B=50	24.47 s 10/10	316.63 s 10/10	37.91 s 10/10	37.60 s 7/10	1797.71 s TL 0/1	1797.55 s TL 0/1	1797.84 s TL 0/1	1797.51 s TL 0/1	1797.77 s TL 0/1

DFO: mean estimated completion time over 9 or 10 starts. GDPopt/MINLP: one measured run per model, including template preparation.

Target attained: final objective ≤ lookup target + tolerance. TL: time limit. †: reported bound conflicts with a saved feasible solution.

GDPopt: BARON + HiGHS (Oct 1). Big-M/Hull: BARON (Sep 25). All on Dantzig; different dates/CPUs. Completion times, not first-hit times.

CSTR5–25 DFO costs include NLP attempts that did not reproduce feasible lookup outcomes; the recorded search targets remain lookup-based.